

Sohan Lele

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EDUCATION

University of Pennsylvania (GRASP Lab) <i>MSE in Robotics</i>	Aug. 2025 – May 2027 <i>Philadelphia, PA</i>
University of California, San Diego <i>BS in Mechanical Engineering, Specialization in Robotics and Controls</i>	Sep. 2021 – Jun. 2025 <i>La Jolla, CA</i>

EXPERIENCE

IRIX tryirix.com <i>Co-Founder & CEO Product Management, Wearable Software Systems</i>	Jan. 2026 – Present <i>Remote</i>
- Leading end-to-end product development of a smart-glasses-based software platform delivering real-time, hands-free personal training - Defined product requirements, system architecture, and feature roadmaps across wearable software, audio interfaces, and context-aware interaction - Designed wearable-first UX optimized for voice, audio, and heads-up displays under latency, power, and usability constraints - Coordinating pilots with hardware partners and gym operators to validate deployment feasibility, reliability, and user experience	
Eversun Energy Inc. <i>Mechanical Engineering Intern Prototyping, Mechanical Design, Ergonomics, Fabrication</i>	Jul. 2024 – Sept. 2024 <i>San Diego, CA</i>
- Directed end-to-end design and fabrication of the Apollo X eTower alpha prototype, taking the system from concept to demo-ready hardware - Redesigned gas-strut lever actuation to eliminate cable fraying and improve ergonomics and reliability; featured in investor demos - Worked cross-functionally with engineering, design, and manufacturing teams to accelerate iteration while maintaining durability and manufacturability	
SoundImaging — MRI Headphones (UCSD Senior Design) <i>Product Design Engineer Medical Devices, Acoustics, Mechanical Design</i>	Jan. 2025 – Jun. 2025 <i>La Jolla, CA</i>
- Designed a pneumatic headphone system enabling clear audio during MRI scans with ambient noise up to 130 dB - Developed modular transducer housings and gasket systems to support rapid iteration and technician usability - Improved signal-to-noise ratio by ~45% through material selection and acoustic testing	

PROJECTS

Steerable Needle Position Estimation — Multi-View Computer Vision (UPenn) <i>Perception Systems Project Medical Robotics, Computer Vision</i>	Aug. 2025 – Dec. 2025 <i>Philadelphia, PA</i>
- Designed a multi-view perception pipeline to estimate the 3D tip position of a steerable needle navigating a reflective gel medium - Integrated segmentation models with classical vision techniques including skeletonization, endpoint detection, and multi-view correspondence - Reconstructed continuous 3D trajectories from synchronized camera views and validated against a deformation model with ~7% relative error - Identified and mitigated failure modes related to glare artifacts, skeleton branching, and frame synchronization	
Ambient AI Clinical Documentation — Human Systems Engineering (UPenn) <i>Systems Design Project AI Systems, Healthcare, Human Factors</i>	Aug. 2025 – Dec. 2025 <i>Philadelphia, PA</i>
- Conducted a human-systems evaluation of an ambient AI clinical documentation system in a pediatric care setting - Performed task analysis and workflow decomposition to identify risks related to automation bias, overreliance, privacy, and clinician trust - Designed a gated, human-in-the-loop workflow enforcing explicit consent, visible system status, and mandatory clinician review - Authored an implementation checklist specifying hardware setup, training, data governance, and feedback loops required for safe scaling	
PrepCaddy — Human-Centered Hardware Design (UPenn Product Design) <i>Product Design Engineer User Research, Prototyping, Fabrication</i>	Aug. 2025 – Dec. 2025 <i>Philadelphia, PA</i>
- Led human-centered design process including pain-point analysis, user interviews, iterative prototyping, and validation - Translated qualitative feedback into mechanical design changes addressing ergonomics, stability, and workflow efficiency - Designed and fabricated a modular hardware system with magnetically-attached components and measured-volume staging - Incorporated manufacturing, material selection, and assembly constraints into final design decisions	

SKILLS

Robotics & Perception: Computer Vision Pipelines, Multi-View Geometry, Sensor Integration, ROS
Mechanical Design & CAD: Fusion 360, SolidWorks, AutoCAD, Rhino 7, ANSYS
Prototyping & Fabrication: 3D Printing, CNC Machining, Laser Cutting, Machine Shop Tools
Programming: Python, MATLAB, Git